

MATH 345 Skeleton Notes

Unit 5: Optimization and training

Section 5.1: Optimization problems and critical points

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Optimization problems

In this unit, we apply multivariate calculus to solve some *optimization problems*.

Definition: Optimization problems Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a scalar-valued function, and consider a set $U \subseteq \mathbb{R}^n$. Then we can consider two optimization problems, in which we call f the

objective function

of the problem, and U the

feasible region

of the optimization problem:

- The

minimization problem

of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *smallest* among all elements of U (so the function is *minimized* on U by $\mathbf{x}^*$).

- The

maximization problem

of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *largest* among all elements of U (so the function is *maximized* on U by $\mathbf{x}^*$).

Example A common optimization problem that we have already encountered in The orthogonal projection onto a subspace involves taking a subspace $U \subseteq \mathbb{R}^n$ and a given vector $\mathbf{x}_0 \in \mathbb{R}^n$, and finding the vector $\mathbf{x}^* \in U$ that is *closest* to $\mathbf{x}_0$, i.e. finding the vector that minimizes the scalar-valued function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ on U , defined by setting $g(\mathbf{x}) = \|\mathbf{x}_0 - \mathbf{x}\|$. Then $\mathbf{x}^*$ is the orthogonal projection of $\mathbf{x}_0$ onto U , and we have seen in Projection theorem that if U has an orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, then

$$\mathbf{x}^* = \frac{\mathbf{x}_0 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\mathbf{x}_0 \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\mathbf{x}_0 \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m.$$

In the section on Least squares from gradients, we return to least squares and derive the normal equations by setting a gradient equal to zero.

Note: Unit 4 connection In Unit 4, projection was a closest-point problem. If W is a subspace and $\mathbf{b}$ is a vector, the projection problem asks for the point $\hat{\mathbf{b}} \in W$ that minimizes

$$\|\mathbf{b} - \mathbf{w}\|$$

over all $\mathbf{w} \in W$. Unit 4 solved this using orthogonality. Unit 5 studies the same kind of question with gradients: minimize a function by looking at how it changes.

Activity: Projection as optimization

Let

$$W = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \subseteq \mathbb{R}^2, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

A general point of W has the form

$$\mathbf{w}(t) = \begin{bmatrix} t \\ 0 \end{bmatrix}.$$

1. Write the squared distance $g(t) = \|\mathbf{b} - \mathbf{w}(t)\|^2$.

2. Minimize $g(t)$.

3. What point of W is closest to $\mathbf{b}$?

4. How does this agree with the projection picture from Unit 4?

Tags. [U5-L01, U4-L04 | C+R | Core]

Critical points

Definition: Minima and maxima Suppose f is a scalar-valued function. A point $\mathbf{a}$ in the domain of f is a

- **local minimum** if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \leq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball, and also in the domain of f . In this case, we say $f(\mathbf{a})$ is a

local minimum value for the function f .

- **local maximum** if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \geq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball, and also in the domain of f . In this case, we say $f(\mathbf{a})$ is a

local maximum value for the function f .

- **global minimum** if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f . In this case, $f(\mathbf{a})$ is the

global minimum value , or

absolute minimum value , of the function f .

- global maximum if $f(\mathbf{a}) \geq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f . In this case, $f(\mathbf{a})$ is the

global maximum value , or

absolute maximum value , of the function f .

A local minimum or local maximum is also called a

local extremum .

Remark The graph of the function $-f$ is obtained by reflecting the graph of the function f in the output axis. This operation turns local and global maxima into local and global minima, and vice versa. So methods for finding minima for a general function often give methods for finding maxima, which makes the two problems often equivalent.

Definition: Critical points Suppose f is a scalar-valued function that is differentiable at a point $\mathbf{a}$. Then we say $\mathbf{a}$ is a

critical point if $\nabla f(\mathbf{a}) = \mathbf{0}$.

For a scalar-valued function $f(x, y)$, a critical point is a solution to the two equations $f_x(x, y) = 0$ and $f_y(x, y) = 0$.

Just as the derivative of a scalar-valued function of a single variable helps identify local minima and maxima, partial derivatives provide information about possible local minima/maxima.

Theorem Suppose f is a scalar-valued function that is differentiable at a point $\mathbf{a}$. If $\mathbf{a}$ is also local extremum of f , then $\mathbf{a}$ is a critical point of f .

Activity

Find the critical points of the function

$$f(x, y) = x^2 + y^2 - 2x + 4y.$$

Activity

Find the critical points of $f(x, y) = x^2 - y^2$.